

TECHNICAL SKILLS:

- Programming languages: JAVA, C++, Python
- Tools: Unit Testing, Machine learning, Stationarity testing (ADF, KPSS), PyTorch, Statistical analysis, NumPy, Pandas, Causality testing, GIT
- Algorithms: Data Structures and Algorithms, Walk forward back testing

EDUCATION:**FORDHAM UNIVERSITY, GABELLI SCHOOL OF BUSINESS**

New York, NY

MS, Quantitative Finance

2024-2026

- Advanced C++, Derivatives, Stochastic Calculus, Monte Carlo, AI in Asset Management, Time series Econometrics, Auto trading systems
- GPA: 3.6 / 4

PUNE INSTITUTE OF COMPUTER TECHNOLOGY (SPPU)

Pune, India

Bachelor of Engineering

2018-2022

- Graduated with a Bachelor of Engineering in Computer Science with an Honors in Data Science with Distinction
- Relevant courses: Object Oriented Programming, Data Base Management Systems, Mathematics, Machine learning, High Performance Computing
- GPA: 3.53 / 4

NEW EIC

Remote, NY

Quant research intern (Part time)

May 2025-Present

- Goal is to trade based on signals generated
- Separate analysis for fundamentally different ETF tickers which represent different sectors like SOXX, XLF, XLV etc.
- Conducted parameter grid optimization for technical indicators across sector ETFs using walk-forward testing, generating preliminary strategy performance metrics
- Moved to using technical indicators like RSI, MACD, Bollinger Bands, Volatility as parameters in time series model like VAR to generate trading signals and back testing them keeping in mind results from preliminary analysis
- The project is in Python

INDEPENDANT STUDY

May 2023-May 2024

Studied mathematics concept to prepare for my master's degree

Real analysis, Probability Theory (Stats 110), Linear Algebra, GMM methods, Stochastic Processes

FLEXTRADE INDIA

Pune, India

Associate Software Developer

February 2023-May 2023

- Merged MarketMaking algo (Order scheduling) and FVPEG algo (Price deciding). Designed a new structure for the project. Migrated the algorithms to use a shared library. Efficiently managed order states
- Debugged order-related issues when merging similar orders for analysis in order analytics code base (BASS)
- Developed client-given tasks in the MarketMaking code base, such as overriding various order-related parameters manually
- The project was in C++

INTERACTIVE BROKERS SOFTWARE SERVICES INDIA, MUMBAI

Mumbai, India

Junior Software Engineer

August 2022-December 2022

- Constructed a Java program in spring to automate storing and generating reports for the different file formats
- Supported existing scalable system project called Control Report Management System. (Java)
- Devised a project to automate complicated RISK factor calculations. Collaborated with the RISK team for smooth development and handling of complicated cases. (Java)

PROJECTS:**COINTEGRATION-BASED TREND STRATEGY RESEARCH**

2025-Present

Built and tested a residual trend-following strategy using cointegrated ETF baskets (e.g., SPY-QQQ-XLK)

- This project developed with the hypothesis that SPY has high influence of tech beta which can be arbitrated after 2015
- Developed a cointegration-based ETF strategy using Johansen VECM on SPY/QQQ/XLK spreads to generate momentum and mean-reverting signals
- Integrated a Random Forest classifier with KMeans regime labeling (using macro features like implied volatility of different expiry and types, Johansen trace statistics which were selected using Granger and DAG) to predict trend/mean-reversion/neutral regimes and avoid poor-performing periods, revealing misalignment of fixed walk-forward windows with actual mean-reversion episodes and prompting a shift toward regime-aware back testing
- Back tested over 2013-2025 using walk-forward framework with realistic 10bps slippage on entries

Code and results are available here: https://github.com/Abhatti-123123/stats_arb/blob/main/README.md**RESEARCH ASSISTANT WITH PROF REN RAW CHEN**

2025-Present

Working on graph-based sentiment analysis

- Knowledge graphs, similarity analysis, graph neural network
- Using news data to build knowledge graph and running neural network over to generate expected price change and price change distributions.
- Based on graph level predictions
- The project is still in exploratory phase because of complexities involved